L'Hôpital's Rule

Can be transformed into $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then use L'Hôpital's Rule

Indeterminate Forms: $\underbrace{\frac{0}{0}, \frac{\infty}{\infty}}_{\text{L'Hôpital's Rule can be applied directly}}, \overbrace{\infty \cdot 0, \infty - \infty}$

Theorem: $f(a) = g(a) = 0$, and f and g are differentiable on an open interval I containing a, and that $g' \neq 0$ on I if $x \neq a$. Then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$$

assuming that the limit on the right side of this equation exists.

Indeterminate Powers: 1^∞, 0^0, ∞^0

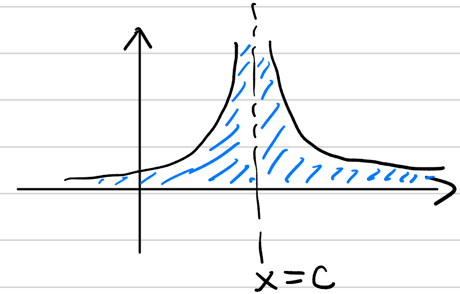
If $\lim_{x \to a} [\ln(f(x))] = L$, then

$$\lim_{x \to a} [f(x)] = \lim_{x \to a} [e^{\ln(f(x))}] = e^L$$

Improper Integrals — Extension of definite Integrals

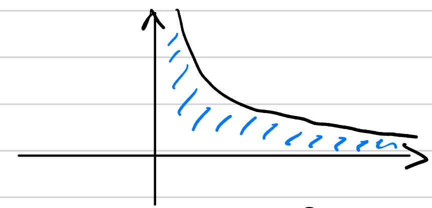
Case 1: Vertical Asymptote:

$$\int_a^b f(x)\,dx = \lim_{d \to c^-}\left[\int_a^d f(x)\,dx\right] + \lim_{g \to c^+}\left[\int_g^b f(x)\,dx\right]$$



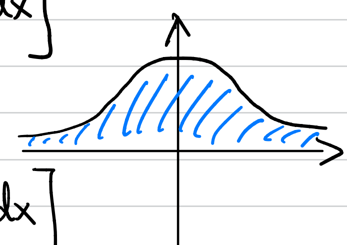
Case 2(a) Infinite Limits of integration.

$$\int_0^\infty f(x)\,dx = \lim_{c \to \infty}\left[\int_0^c f(x)\,dx\right]$$



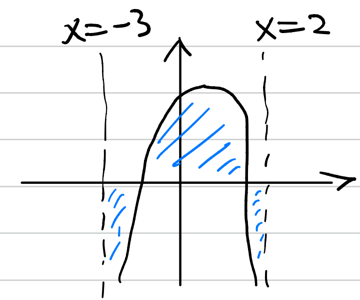
Case 2(b)

$$\int_{-\infty}^{\infty} f(x)\,dx = \lim_{c \to -\infty}\left[\int_c^5 f(x)\,dx\right] + \lim_{d \to \infty}\left[\int_5^d f(x)\,dx\right]$$



Case 3: Two Vertical Asymptote

$$\int_{-3}^{2} f(x)\,dx = \lim_{c \to 3^+}\left[\int_c^0 f(x)\,dx\right] + \lim_{d \to 2^-}\left[\int_0^d f(x)\,dx\right]$$



If any of the limit gives you infinity, the integral diverges

Infinite Sequences

An expression about n

$\{a_n\} = \{a_1, a_2, \cdots\}$.

starts from $n=1$

Monotonic:
① increasing
② decreasing
③ constant

lower bound (L):
for all n, $L \leq a_n$.

upper bound (U):
for all n, $U \geq a_n$

greatest lower bound (g.l.b.):
for all lower bounds L, g.l.b. $\geq L$

least upper bound (l.u.b.):
for all upper bounds U, l.u.b $\leq U$.

DEFINITIONS The sequence $\{a_n\}$ **converges** to the number L if for every positive number ε there corresponds an integer N such that

$$|a_n - L| < \varepsilon \qquad \text{whenever} \qquad n > N.$$

If no such number L exists, we say that $\{a_n\}$ **diverges**.

If $\{a_n\}$ converges to L, we write $\lim_{n\to\infty} a_n = L$, or simply $a_n \to L$, and call L the **limit** of the sequence (Figure 10.2).

DEFINITION The sequence $\{a_n\}$ **diverges to infinity** if for every number M there is an integer N such that for all n larger than N, $a_n > M$. If this condition holds we write

$$\lim_{n\to\infty} a_n = \infty \qquad \text{or} \qquad a_n \to \infty.$$

Similarly, if for every number m there is an integer N such that for all $n > N$ we have $a_n < m$, then we say $\{a_n\}$ **diverges to negative infinity** and write

$$\lim_{n\to\infty} a_n = -\infty \qquad \text{or} \qquad a_n \to -\infty.$$

Infinite Series : the sum of an infinite sequence of numbers

DEFINITIONS Given a sequence of numbers $\{a_n\}$, an expression of the form

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots$$

is an **infinite series**. The number a_n is the ***n*th term** of the series. The sequence $\{s_n\}$ defined by

$$\begin{aligned} s_1 &= a_1 \\ s_2 &= a_1 + a_2 \\ &\vdots \\ s_n &= a_1 + a_2 + \cdots + a_n = \sum_{k=1}^{n} a_k \\ &\vdots \end{aligned}$$

is the **sequence of partial sums** of the series, the number s_n being the ***n*th partial sum**. If the sequence of partial sums converges to a limit L, we say that the series **converges** and that its **sum** is L. In this case, we also write

$$a_1 + a_2 + \cdots + a_n + \cdots = \sum_{n=1}^{\infty} a_n = L.$$

If the sequence of partial sums of the series does not converge, we say that the series **diverges**.

If $|r| < 1$, the geometric series $a + ar + ar^2 + \cdots + ar^{n-1} + \cdots$ converges to $a/(1-r)$:

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}, \qquad |r| < 1.$$

If $|r| \geq 1$, the series diverges.

THEOREM 7 If $\sum_{n=1}^{\infty} a_n$ converges, then $a_n \to 0$.

The *n*th-Term Test for Divergence

$\sum_{n=1}^{\infty} a_n$ diverges if $\lim_{n\to\infty} a_n$ fails to exist or is different from zero.

THEOREM 8 If $\sum a_n = A$ and $\sum b_n = B$ are convergent series, then

1. *Sum Rule:* $\sum(a_n + b_n) = \sum a_n + \sum b_n = A + B$
2. *Difference Rule:* $\sum(a_n - b_n) = \sum a_n - \sum b_n = A - B$
3. *Constant Multiple Rule:* $\sum ka_n = k\sum a_n = kA$ (any number k).

THEOREM 7 If $\sum_{n=1}^{\infty} a_n$ converges, then $a_n \to 0$.

$\sum_{n=1}^{\infty} a_n$ converges $\not\Leftarrow\Rightarrow$ $a_n \to 0$

The nth-Term Test for Divergence

$\sum_{n=1}^{\infty} a_n$ diverges if $\lim_{n\to\infty} a_n$ fails to exist or is different from zero.

$a_n \to L \neq 0 \Rightarrow \sum_{n=1}^{\infty} a_n$ diverges

3. Determine if each statement below is always true or sometimes false. **If the statement is false, provide a counterexample.**

(a) If $\lim_{n\to\infty} a_n = 0$, then the series $\sum a_n$ converges. False counterexample: $\sum_{k=1}^{\infty}\left(\frac{1}{n}\right)$ Harmonic Series

(b) If $\lim_{n\to\infty} a_n = 0$, then the sequence $\{a_n\}$ converges. True

(c) The series $\sum_{n=0}^{\infty} \frac{1}{n+1}$ diverges. True $\sum_{n=0}^{\infty}\left(\frac{1}{n+1}\right) = \sum_{n=1}^{\infty}\left(\frac{1}{n}\right)$

(d) The sum of two divergent series also diverges. False counterexample: $\sum_{k=0}^{\infty}\left[(-1)^k + (-1)^{k+1}\right]$

(e) The series $\sum_{n=1}^{\infty} r^n$ converges to $\frac{1}{1-r}$ if $|r| < 1$. False

$$\sum_{n=0}^{\infty}(r^n) = \frac{1}{1-r}$$

$$\sum_{n=1}^{\infty}(r^n) = \frac{1}{1-r} - 1$$

The Integral Test

THEOREM 9—The Integral Test
Let $\{a_n\}$ be a sequence of positive terms. Suppose that $a_n = f(n)$, where f is a continuous, positive, decreasing function of x for all $x \geq N$ (N a positive integer). Then the series $\sum_{n=N}^{\infty} a_n$ and the integral $\int_N^{\infty} f(x)\,dx$ both converge or both diverge.

Convergence Tests that we have learned so far :

<u>Test for convergence</u> : ① Definition (partial sums, telescoping series)
② special series (geometric series w/ $|r|<1$, p-series w/ $p>1$)
③ Integral test (only for continuous, positive, decreasing "functions")
④ Basic comparison test & limit comparison test.

<u>Test for divergence</u> : ① Definition (partial sums)
② special series (geometric series w/ $|r|\geq 1$, harmonic series $\sum_{k=1}^{\infty}\frac{1}{k}$, $\sum_{k=1}^{\infty}(-1)^k$, p-series w/ $p\leq 1$)
③ n-th term test ($\lim_{n\to\infty} a_n \neq 0$)
④ Integral test (only for continuous, positive, decreasing "functions")
⑤ Basic comparison test & limit comparison test.

<u>p-series</u> $\sum_{k=1}^{\infty}\left(\frac{1}{k^p}\right)$ is called p-series.

when $p>1$, the p-series converges.
when $p=1$, this is the harmonic series, therefore diverges.
when $p<1$, the p-series diverges.

Comparison and Limit Comparison Test

(Basic)

THEOREM 10—Direct Comparison Test

Let $\sum a_n$ and $\sum b_n$ be two series with $0 \leq a_n \leq b_n$ for all n. Then

1. If $\sum b_n$ converges, then $\sum a_n$ also converges.
2. If $\sum a_n$ diverges, then $\sum b_n$ also diverges.

THEOREM 11—Limit Comparison Test

Suppose that $a_n > 0$ and $b_n > 0$ for all $n \geq N$ (N an integer).

1. If $\displaystyle\lim_{n\to\infty} \frac{a_n}{b_n} = c$ and $c > 0$, then $\sum a_n$ and $\sum b_n$ both converge or both diverge.
2. If $\displaystyle\lim_{n\to\infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.
3. If $\displaystyle\lim_{n\to\infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Convergence Tests that we have learned so far:

Test for convergence:
① Definition (partial sums, telescoping series)
② special series (geometric series w/ $|r|<1$, p-series w/ $p>1$)
③ Integral test (only for continuous, positive, decreasing "functions")
④ Basic comparison test & limit comparison test.
⑤ Ratio test w/ $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|<1$
⑥ Root test w/ $\lim_{n\to\infty}\left[\sqrt[n]{|a_n|}\right]<1$

Test for divergence:
① Definition (partial sums)
② special series (geometric series w/ $|r|\geq 1$, harmonic series $\sum_{k=1}^{\infty}\frac{1}{k}$, $\sum_{k=1}^{\infty}(-1)^k$, p-series w/ $p\leq 1$)
③ n-th term test ($\lim_{n\to\infty} a_n \neq 0$)
④ Integral test (only for continuous, positive, decreasing "functions")
⑤ Basic comparison test & limit comparison test.
⑥ Ratio test w/ $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|>1$
⑦ Root test w/ $\lim_{n\to\infty}\left[\sqrt[n]{|a_n|}\right]>1$

Ratio and Root Tests

DEFINITION A series $\sum a_n$ **converges absolutely** (is **absolutely convergent**) if the corresponding series of absolute values, $\sum |a_n|$, converges.

THEOREM 12—The Absolute Convergence Test
If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

$\sum_{n=1}^{\infty} |a_n|$ converges $\Rightarrow \sum_{n=1}^{\infty} a_n$ converges

THEOREM 13—The Ratio Test
Let $\sum a_n$ be any series and suppose that

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \rho.$$

$\Rightarrow$ converges

Then **(a)** the series *converges absolutely* if $\rho < 1$, **(b)** the series *diverges* if $\rho > 1$ or ρ is infinite, **(c)** the test is *inconclusive* if $\rho = 1$.

THEOREM 14—The Root Test
Let $\sum a_n$ be any series and suppose that

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \rho.$$

$\Rightarrow$ converges

Then **(a)** the series *converges absolutely* if $\rho < 1$, **(b)** the series *diverges* if $\rho > 1$ or ρ is infinite, **(c)** the test is *inconclusive* if $\rho = 1$.

Convergence Tests that we have learned so far:

Test for convergence:
1. Definition (partial sums, telescoping series)
2. special series (geometric series w/ $|r|<1$, p-series w/ $p>1$)
3. Integral test (only for continuous, positive, decreasing "functions")
4. Basic comparison test & limit comparison test.
5. Ratio test w/ $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|<1$
6. Root test w/ $\lim_{n\to\infty}\left[\sqrt[n]{|a_n|}\right]<1$
7. Alternating test (for alternating series).

Test for divergence:
1. Definition (partial sums)
2. special series (geometric series w/ $|r|\geq 1$, harmonic series $\sum_{k=1}^{\infty}\frac{1}{k}$, $\sum_{k=1}^{\infty}(-1)^k$, p-series w/ $p\leq 1$)
3. n-th term test ($\lim_{n\to\infty} a_n \neq 0$)
4. Integral test (only for continuous, positive, decreasing "functions")
5. Basic comparison test & limit comparison test.
6. Ratio test w/ $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|>1$
7. Root test w/ $\lim_{n\to\infty}\left[\sqrt[n]{|a_n|}\right]>1$

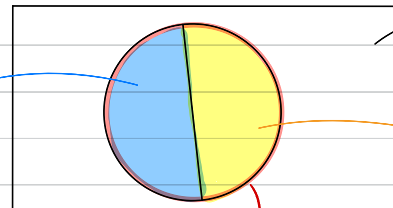
Alternating Series

Let $u_n = |a_n|$

DEFINITION A series that is convergent but $\boxed{\text{not}}$ absolutely convergent is called **conditionally convergent**.

$\sum_{n=1}^{\infty} a_n$ diverges

$\sum_{n=1}^{\infty} a_n$ converges absolutely $\Leftrightarrow \sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty} |a_n|$ converges



$\sum_{n=1}^{\infty} a_n$ converges conditionally.

$\sum_{n=1}^{\infty} a_n$ converges

THEOREM 15—The Alternating Series Test

The series

$$\sum_{n=1}^{\infty} (-1)^{n+1} u_n = u_1 - u_2 + u_3 - u_4 + \cdots$$

converges if the following conditions are satisfied:

1. The u_n's are all positive.
2. The u_n's are eventually nonincreasing: $u_n \geq u_{n+1}$ for all $n \geq N$, for some integer N.
3. $u_n \longrightarrow 0$.